removes the attempted candidate while keeping the task active. **Region-level exhaustion** invokes alternative room–functional-region hypotheses grounded in the hierarchy; these hypotheses become geometric observation regions and are ranked using semantic relevance, coverage, and travel cost. A **task-level negative outcome** is produced when the task-specific stopping condition is reached, after which CPTG updates downstream logic. Within semantic recovery, newly covered portions of an unproductive region reduce its belief, shifting search toward alternative hypotheses.

Novel detections are accumulated online through spatially consistent multi-view fusion and materialized in the updated scene graph after execution. Thus, task states and semantic-region beliefs can affect replanning within the current execution, whereas newly created object instances support later use of the revised map.

After each terminal verification, the system updates the task state, candidate poses, and semantic search prior, recomputes $\mathcal{E}_t$, and invokes JTVP from the current robot pose. When a task reaches a task-level outcome, CPTG additionally updates $\mathcal{R}_t$ and evaluates $\mathcal{R}_{\text{cond}}$ to activate or skip downstream conditional tasks. This closes the loop among task logic, 3D observation, geometric motion, and online semantic evidence.

G. Methodology Summary

LogicFly presents a hierarchical perception–reasoning–motion framework for long-horizon aerial navigation. It first builds a semantic–geometric representation that grounds object evidence and room structure in observed geometry. Natural-language instructions are then organized into a Conditional Partial-Order Task Graph (CPTG), enabling precedence constraints and observation-dependent branches to be reasoned about jointly with task-conditioned viewpoints. Within a receding-horizon planning scheme, the system coordinates task selection, viewpoint generation, and collision-aware motion, while safe-corridor-constrained MINCO optimization converts the selected discrete route into a dynamically feasible trajectory. Terminal observations close the loop by verifying task outcomes and updating semantic evidence, candidate viewpoints, and task states. Once a task reaches its terminal outcome, the updated semantic confidence is used to revise the semantic map by incorporating newly verified instances and the corresponding scene-graph updates. Through this integration of semantic task reasoning and geometric motion generation, LogicFly supports reliable execution of complex multi-target instructions in unknown indoor environments.

V. Experiments

We evaluate LogicFly through simulation, ablation studies, and real-world UAV experiments, covering benchmark performance, the contributions of CPTG and JTVP, and practical deployment.

A. Experimental Setup

Evaluation benchmark. We evaluate all methods on six HM3D [40] scenes: 00070, 00166, 00273, 00290, 00337, and 00409. The benchmark contains 152 language instructions, including 61 short single-target instructions, 60 long-horizon instructions confined to a single floor, and 31 long-horizon cross-floor instructions. Using the valid initial-state seeds and two repeated trials, each method is evaluated over 854 episodes. The benchmark covers ordered execution, conditional branches, target absence, failure recovery, spatial relations, and instance disambiguation.

Evaluation metrics. We report task Success Rate (SR), Success weighted by Path Length (SPL), and Completion Time (CT). An episode is considered successful only when all required task outcomes are completed while satisfying the instruction-level ordering, branching, spatial, recovery, and disambiguation constraints. Reaching the correct objects without satisfying the complete instruction is counted as a failure. For episode e, SPL is computed as

$$\text{SPL}_e = S_e \frac{L_e^{\text{ref}}}{\max(L_e^{\text{ref}}, L_e)}, \tag{15}$$

where S_e is the strict binary task-success indicator, L_e is the executed trajectory length, and L_e^{ref} is the shortest feasible path length under the same task constraints. SPL is not applicable to target-absence episodes.

Baselines. We compare LogicFly against four representative map-based and spatial-reasoning navigation methods. OneMap [22] is a zero-shot multi-object navigation method that constructs a real-time open-vocabulary feature map and queries the unified map to locate multiple target categories. MapGPT [6] is an LLM-based VLN method that incorporates an environment map into the language prompt and adaptively updates its planned route according to the explored topology. SpatialGPT [7] performs zero-shot VLN by maintaining a structured spatial memory and applying spatial chain-of-thought reasoning to select navigation actions. SpatialNav [28] represents perceived objects and their spatial relations using a spatial scene graph, which supports graph-conditioned reasoning for zero-shot navigation.

Implementation details. LogicFly first compiles each instruction into a conditional and precedence-constrained task graph, which is subsequently executed by the proposed task-graph runner. Reusable semantic maps are constructed from registered RGB-D observations. The mapping front end uses SAM 3.1 in image mode with a fixed vocabulary of 30 indoor object categories and a confidence threshold of 0.50. Video tracking is disabled. Boxer lifts the accepted masks into 3-D and performs multi-view instance fusion using an IoU threshold of 0.20, an instance-confidence threshold of 0.45, and a minimum of four observations. During task execution, online object evidence is obtained from calibrated RGB-D observations using open-vocabulary recognition and is projected into the world frame using the estimated camera pose and aligned depth.

The joint task-and-viewpoint planner considers a maximum of three candidate viewpoints and uses a receding planning horizon of $H = 3$ tasks. The nominal horizontal velocity, vertical velocity, and yaw rate are $1.0\,\text{m/s}$, $0.5\,\text{m/s}$, and $1.0\,\text{rad/s}$, respectively. A collision safety margin of $0.10\,\text{m}$ is enforced during trajectory generation in both simulated

TABLE I

BENCHMARK RESULTS GROUPED BY TASK DIFFICULTY AND FLOOR SCOPE. THE CLEAN SCHEDULE CONTAINS 270 SINGLE-FLOOR AND 38 CROSS-FLOOR SINGLE-TARGET EPISODES, AND 364 SINGLE-FLOOR AND 182 CROSS-FLOOR MULTI-TARGET EPISODES. SR AND SPL ARE PERCENTAGES; CT IS AVERAGED OVER SUCCESSFUL EPISODES. A DASH INDICATES NO SUCCESSFUL EPISODE. ONEMAP USES ITS ORDERED-CATEGORY PROJECTION. BOLD DENOTES THE BEST RESULT IN EACH TASK GROUP.

Task difficulty	Method	Single-floor			Cross-floor		
		SR↑ (%)	SPL↑ (%)	CT↓ (s)	SR↑ (%)	SPL↑ (%)	CT↓ (s)
Single-target	SpatialNav	8.15	5.89	22.35	–	–	–
	SpatialGPT	10.37	2.69	115.39	–	–	–
	MapGPT	27.78	14.35	42.95	18.42	13.08	81.57
	OneMap	52.59	35.45	77.32	21.05	20.30	49.88
	LogicFly	**97.78**	**87.28**	**8.19**	**94.74**	**86.03**	**8.99**
Multi-target	MapGPT	1.10	0.73	96.02	–	–	–
	OneMap	12.36	8.16	150.13	6.59	6.59	90.59
	LogicFly	**96.70**	**82.49**	**10.33**	**96.70**	**81.45**	**14.70**

and real-world experiments. All language-model components use Qwen3.8-Flash. All perception, planning, and recovery parameters are fixed across scenes and episode repetitions.

Runtime protocol. Each evaluation process is subject to a fixed time limit of 240 s. A timeout or execution error is counted as a failed episode. For LogicFly, CT is measured using its native completion timestamp and excludes one-time map preparation. For the comparison methods, CT is measured from process startup to task completion. CT is averaged over successful episodes only.

B. Comparison with Baseline Methods in Simulation

We compare LogicFly with HM3D adaptations of OneMap [22], MapGPT [6], SpatialGPT [7], and SpatialNav [28]. All methods run on the same 854-episode clean schedule. OneMap retains its native ordered-category projection, which omits relations, conditional branches, verified absence, recovery, and return constraints. All baselines and LogicFly are evaluated using the same frozen strict full-task evaluator. SpatialGPT and SpatialNav do not successfully complete any cross-floor episode, resulting in an SR of 0%; therefore, their remaining cross-floor metrics are not reported.

As shown in Table I, LogicFly consistently outperforms the compared methods across target count and floor scope. On single-target tasks, it reaches 97.78% SR, 87.28% SPL, and 8.19 s CT on single-floor episodes, and 94.74% SR, 86.03% SPL, and 8.99 s CT on cross-floor episodes. On the harder multi-target tasks, it maintains a 96.70% SR on both floor scopes, with SPL values of 82.49% and 81.45% and CT values of 10.33 s and 14.70 s for single-floor and cross-floor episodes, respectively. SpatialGPT and SpatialNav have no successful multi-target episodes and are therefore omitted from that block. Across the full 854-episode schedule, LogicFly completes 828 episodes for a 96.96% strict full-task SR and an overall SPL of 0.8394.

Figure 8 presents a qualitative execution example in a three-floor Habitat environment. LogicFly completes the cross-floor route and all required subtasks, with terminal views providing visual evidence for each successful outcome.

All baselines retain their native planning, action, and stopping interfaces. OneMap receives the protocol-defined ordered-category projection, whereas MapGPT, SpatialGPT, and SpatialNav receive the frozen task specification through their native interfaces. Timeouts and algorithm failures remain unsuccessful terminal outcomes.

C. Ablation Studies

TABLE II

ABLATION RESULTS ON THE MULTI-TARGET SUBSET OF 546 EPISODES. SR AND SPL ARE PERCENTAGES, AND CT IS AVERAGED OVER SUCCESSFUL EPISODES.

Variant	SR↑ (%)	SPL↑ (%)	CT↓ (s)
w/o CPTG	64.84	61.85	35.63
CPTG + fixed views	**96.70**	78.03	11.65
CPTG + JTVP	**96.70**	**82.15**	**6.39**

We evaluate CPTG and JTVP on the 546 multi-target episodes from the frozen six-scene HM3D schedule. This subset includes ordered, freely schedulable, conditional-branch, verified-absence, semantic-recovery, spatial-relation, cross-floor, and target-disambiguation tasks. Running all three variants yields 1638 end-to-end multi-target trials.

All variants are executed end to end using the native Habitat runner, online open-vocabulary verification, and the same strict evaluator. They share identical scene representations, start poses, motion constraints, termination rules, and evaluation settings. *w/o CPTG* removes task dependencies, conditional rules, and task-conditioned candidate generation, retaining only target labels, coarse room/floor scopes, generic observation constraints, and fixed language order. *CPTG + fixed views* retains CPTG and task-order optimization but uses one independently selected fixed observation pose per task. *CPTG + JTVP* jointly optimizes task ordering and observation viewpoints. The evaluator requires every required outcome and effective order, forbids visits to skipped branches, and checks spatial/reference constraints.

As shown in Table II, removing CPTG reduces multi-target SR to 64.84% and SPL to 61.85%, while increasing successful-task CT to 35.63 s. Without an explicit task graph, the system cannot reliably enforce precedence relations, activate observation-dependent branches, or distinguish required



Fig. 8. Examples of long-instruction execution in Habitat. Each row pairs a natural-language instruction with the corresponding sequence of first-person observations, covering freely ordered subtasks and observation-dependent search or verification.

tasks from skipped alternatives. This leads to incorrect execution orders, unnecessary visits, and repeated recovery attempts, demonstrating that CPTG is the principal contributor to full-task correctness. Restoring CPTG with fixed viewpoints raises SR by 31.86 percentage points to 96.70%, increases SPL to 78.03%, and reduces CT to 11.65 s. These improvements show that explicit task dependencies and conditional transitions eliminate most logical execution failures and substantially reduce redundant motion. Replacing the independently selected fixed viewpoints with JTVP preserves the same 96.70% SR while further increasing SPL by 4.12 percentage points to 82.15% and reducing CT by 5.26 s to 6.39 s. Thus, CPTG primarily ensures correct execution of the complete instruction, whereas JTVP improves efficiency by jointly selecting task order and observation poses according to travel and terminal-view costs. The 30 timeout or infrastructure-failure outcomes of w/o CPTG remain in the 546-episode denominator.

D. Evaluation in Real-World Environments

Experimental platform. We conduct real-world experiments using a custom-built quadrotor equipped with an NVIDIA Jetson Orin NX (16 GB), a PX4 flight controller, a Livox Mid-360 LiDAR, and an Intel RealSense D435 depth camera. The D435 provides synchronized 640×480 RGB and depth observations for online verification and 3D projection. FAST-LIO [41] provides onboard localization, while LiDAR-based occupancy mapping supports real-time trajectory optimization. The UAV is constrained to a maximum linear velocity of 1.0 m/s, linear acceleration of 1.0 m/s^2, and angular velocity of 1.0 rad/s.

As illustrated in Fig. 1, LogicFly completes a four-task instruction end to end with a physical UAV in a real multi-room environment. After the laptop is not found at its instructed location, the UAV performs semantic recovery in the reading room, then verifies the laptop, toolbox, microwave state, and water bottle. The map panel provides a bird's-eye visualization of the executed route, while numbered views document the recovery and verification sequence.



Fig. 9. Representative flight observations and visual detections from the confined indoor home deployment.

Figures 9–11 show representative observations from the confined indoor home and two office deployments.

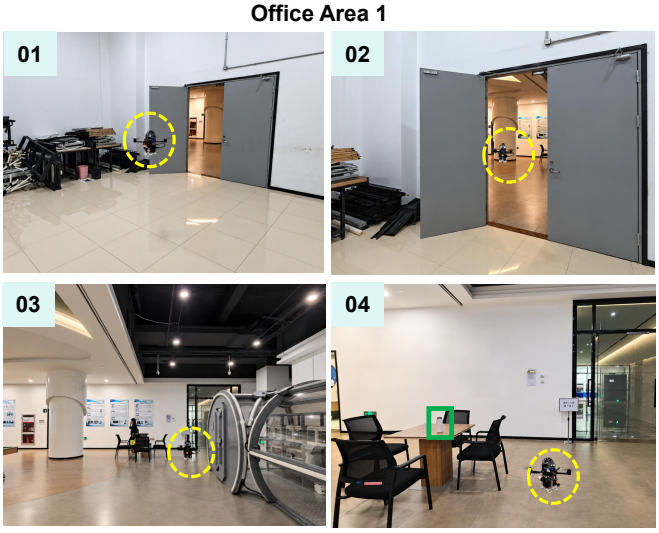


Fig. 10. Representative flight observations and visual detections from Office Area 1.

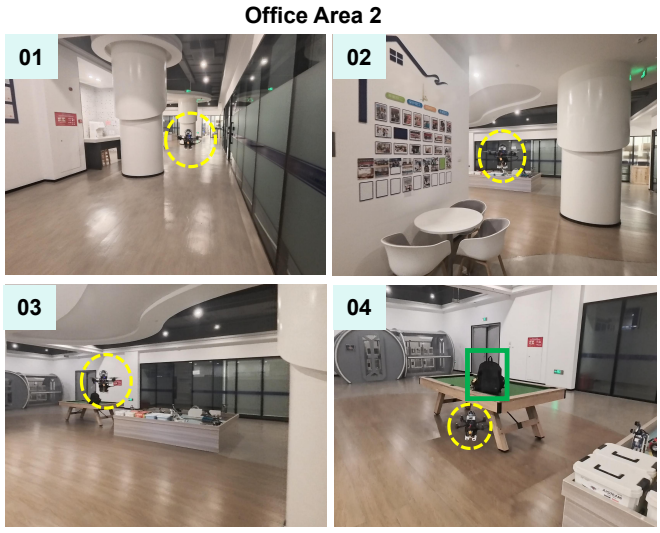


Fig. 11. Representative flight observations and visual detections from Office Area 2.